# Glymmy

## Your glycemic companion — AI health platform

*(glycemic + my — a unified health data platform)*

Glymmy connects CGM, wearables, food photos, and lab reports into one AI-powered health companion. It uses the multimodal **Gemma 4** model to turn raw clinical data into personalized, actionable insights — running locally by default, with zero paid dependencies.

**Built for the Harvard HSIL Hackathon 2026.**

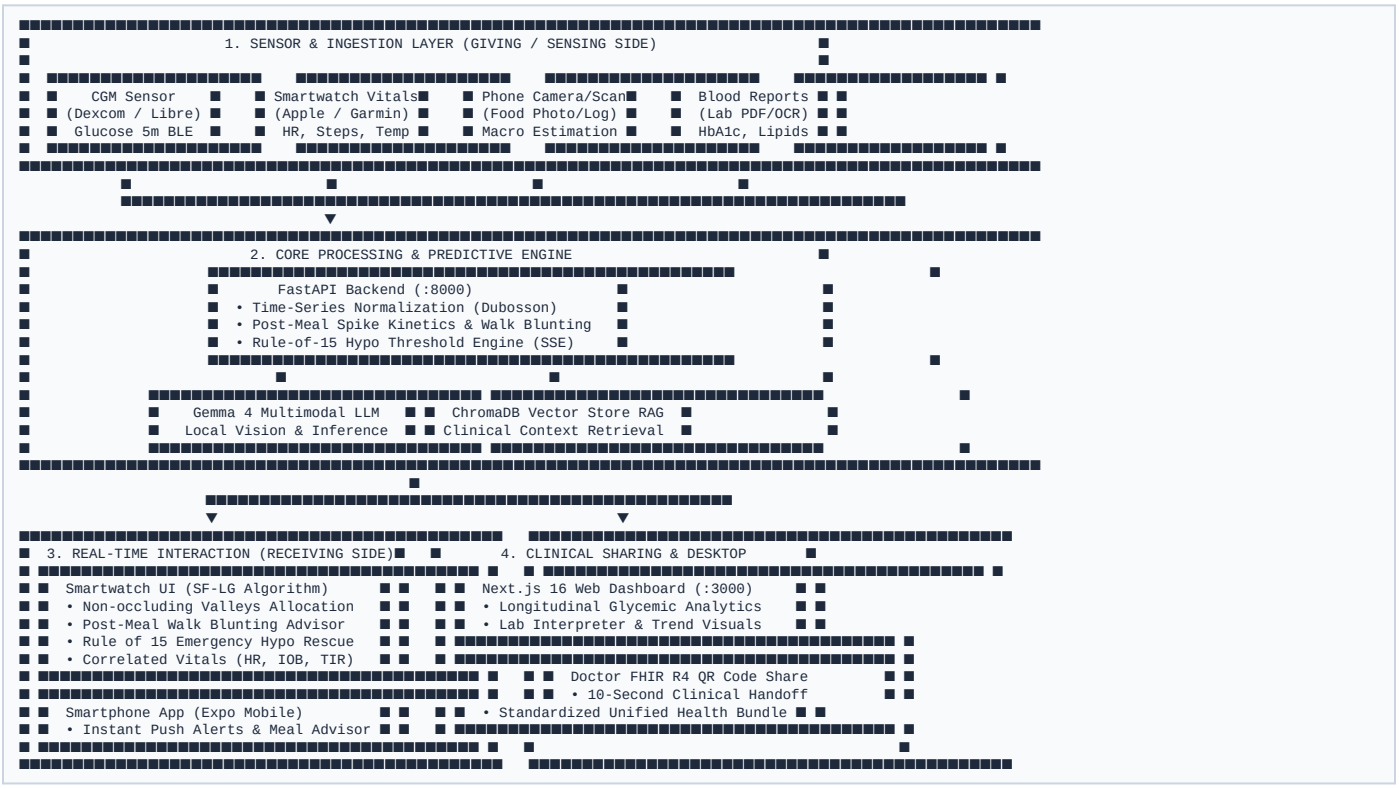
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## What it does

| Feature | Description |
|---|---|
| **CGM Dashboard** | Real-time glucose time-series, time-in-range (TIR), hypo/hyper alerts |
| **Smartwatch UI (SF-LG)** | Circular 300x300 watch interface with **Space-Filling Line Graph (SF-LG)** greedy space allocation for post-meal walk advice & vitals |
| **Food Scanner** | Photo or text → full nutrition breakdown + AI post-prandial glucose spike prediction |
| **Lab Interpreter** | AI interpretation of blood reports with risk assessment |
| **Wearable Sync** | CGM, Apple Watch, Garmin HR monitor, Fitbit activity tracker (HR, steps, sleep, SpO2) |
| **RAG Chat** | Ask questions about your health data with cited clinical sources |
| **FHIR QR Share** | Generate a QR code containing a FHIR bundle for doctor visits in 10 seconds |
| **Real-time Alerts** | Multi-level alerts (Rule of 15 rescue protocol) & SSE push notifications |

## System Architecture: Bidirectional Data Flow

Glymmy operates on a closed-loop feedback architecture: devices act as continuous physiological emitters (**Giving Side**), while the local AI engine processes correlations and delivers actionable micro-interventions back to the patient and clinician (**Receiving & Acting Side**).




## Smartwatch Feature: SF-LG Greedy Space Allocation

Small circular smartwatches (300×300px) present a severe UI challenge: standard cards or overlay complications occlude the glucose curve or get clipped by circular borders.

Glymmy implements the **SF-LG (Space-Filling Line Graph)** algorithm to solve this:
• **Dynamic Valleys Discovery:** Scans the active CGM time-series curve to identify empty geometric headroom ('valleys above') and footroom ('valleys below').
• **Greedy Area Allocation:** Sorts candidate rectangular pockets by area (width × height) and allocates the largest non-overlapping spaces to contextual micro-widgets.




***Figure 1:*** *The SF-LG algorithm dynamically hosting the Post-Meal Spike & Walk Advisor in the empty valley above the CGM curve. Centered complications (TIR 84%, 10:09, IOB 1.4U) contour within the circular bezel.*

# Interactive Smartwatch Scenarios Showcase

### ■ S4: Meal Spike & Walk (Signature)




**Post-Meal Walk Advisor:** Predicts peak (192 mg/dL); tapping *Simulate 15-min Walk* flattens curve in real time to 138 mg/dL (-25 mg/dL).

### ■ S1: Point Vitals Correlation



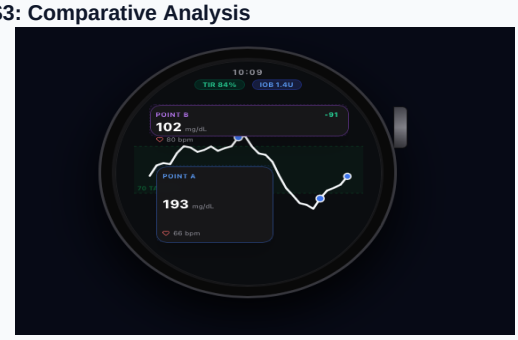

**Point Vitals:** Tap any point to surface correlated Heart Rate (bpm) and active Insulin on Board (IOB) from wearable fusion.

### ■■ S2: Rule of 15 Hypo Protocol



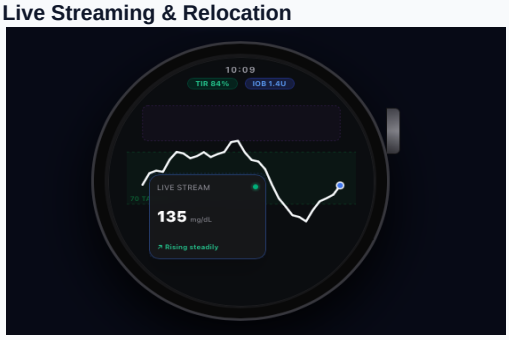

**Rule of 15 Rescue:** Ambient pulsing alert expands into clinical guidance (eat 15g fast carbs & retest in 15m) with tachycardia monitoring.

### ■■ S3: Comparative Analysis




**Comparative Analysis:** Tap two points to compare exact glycemic delta (Δ mg/dL) and heart rate variations.

### ■ S5: Live Streaming & Relocation




**Live Streaming:** Live simulation where widgets smoothly glide and reposition as glycemic peaks and valleys shift.